Tutorial Week 5-6

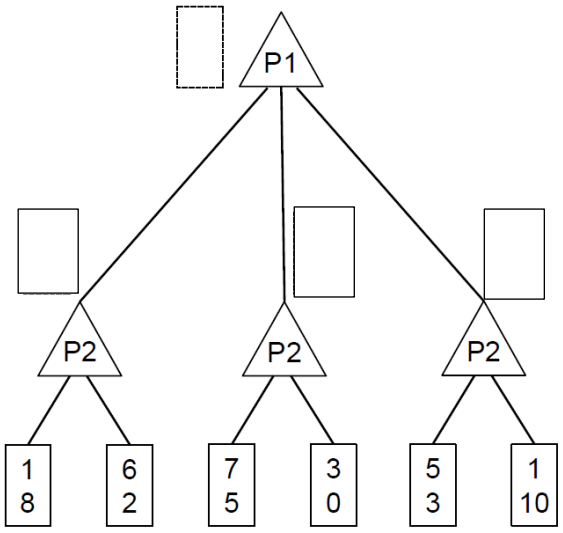
This weeks tutorial will cover both Adversarial Games and Knowledge representation and reasoning.

- Review definitions.
- Game trees and alpha-beta pruning.
- Explore the minimax and alpha-beta pruning algorithms.
- Backward/forward chaining
- Resolution example

1) Define in your own words the following terms: zero sum game, game tree, pruning, terminal state, strategy, minimax value, utility function.

2) Examine the following game tree description and answer the questions below.

For the following game tree, each player maximizes their respective utility. Let x, y respectively denote the top and bottom values in a node. Player 1 uses the utility function $U_1(x, y) = x$.



Both players know that Player 2 uses the utility function $U_2(x, y) = x - y$.

a) Fill in the rectangles in the figure with the pair of values returned by each max node.

b) On the game tree above, put an 'X' on branches that do not need to be explored or write 'None'. Assume the branches are explored from left to right.

Knowledge representation and reasoning

3) Define in your own words the following terms: Logic, inference, model, satisfiability, validity, completeness, entailment, propositional logic, first-order logic, resolution, existential/universal quantification, forward/backward chaining, declarative vs procedural.

4) Convert the following set of sentences to clausal form.

S1: $A \Leftrightarrow (B \vee E)$.

S2: $E \Rightarrow D$.

S3: $C \wedge F \Rightarrow \neg B$.

S4: $E \Rightarrow B$.

S5: $B \Rightarrow F$.

S6: $B \Rightarrow C$

5) Resolution.

Consider the following sentence:

$[(\text{Study} \Rightarrow \text{Pass}) \vee (\text{Persistence} \Rightarrow \text{Pass})] \Rightarrow [(\text{Study} \wedge \text{Persistence}) \Rightarrow \text{Pass}]$

a) Use truth tables to determine whether this sentence is valid, satisfiable, or unsatisfiable.
b) Convert the left-hand and right-hand sides of the main implication into CNF, showing each step, and explain how the results confirm your answer to (a)
c) Prove your answer to (a) using resolution.
d) Is there a contradiction in the clauses of question 4?

Optional: You may explore the following notebook files

6) **Download the Tut5search.ipynb file. Start jupyter lab and work through the notebook.**

End